

4. Haemoglobin exhibits quaternary structure and consists of four subunits, two subunits called alpha-globin and two subunits called beta-globin.

(a) Define quaternary structure.

[1]

- (b) (i) The beta-globin molecule consists of 146 amino acids. State the minimum number of mRNA nucleotides required to code for this molecule.

[1]

- (ii) The genetic code is a triplet code and is degenerate. The table below shows the **mRNA codons**, that code for specific amino acids.

		Second letter				
		U	C	A	G	
First letter	U	UUU Phenyl-alanine UUC UUA Leucine UUG	UCU Serine UCC UCA UCG	UAU Tyrosine UAC UAA Stop codon UAG Stop codon	UGU Cysteine UGC UGA Stop codon UGG Tryptophan	Third letter U C A G
	C	CUU Leucine CUC CUA CUG	CCU Proline CCC CCA CCG	CAU Histidine CAC CAA Glutamine CAG	CGU Arginine CGC CGA CGG	
	A	AUU Isoleucine AUC AUA AUG Methionine initiation codon	ACU Threonine ACC ACA ACG	AAU Asparagine AAC AAA Lysine AAG	AGU Serine AGC AGA Arginine AGG	
	G	GUU Valine GUC GUA GUG	GCU Alanine GCC GCA GCG	GAU Aspartic acid GAC GAA Glutamic acid GAG	GGU Glycine GGC GGA GGG	

Use the table to explain why it is necessary that the code is a triplet code and why the code is said to be degenerate.

[2]



- (iii) The table below shows part of the DNA sequence that codes for beta-globin and the corresponding mRNA codons, tRNA anticodons and the amino acid sequence of the molecule.

Complete the table.

[4]

	Codon position number					
	3	4	5	6	7	8
DNA coding strand	GAC			CTC	CTC	
mRNA codons		ACU		GAG	GAG	
tRNA anticodons			GGA	CUC	CUC	
Amino acid sequence				Glutamic acid	Glutamic acid	Lysine

- (c) Describe the events which take place in the production of the beta-globin molecule after the ribosome has attached to the mRNA. [5]

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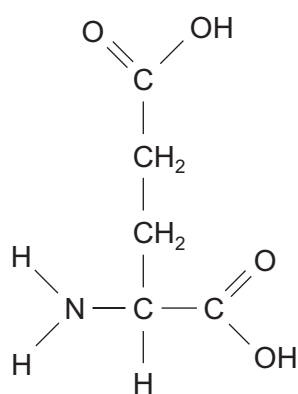
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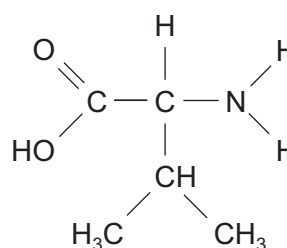


- (d) A mutation, where a single base was substituted by a different base, resulted in glutamic acid being replaced by valine at position 6 in the beta-globin molecule. Use the information provided to determine the mutated DNA base sequence at position 6 on the coding strand of DNA. [1]

- (e) The diagrams below show the structures of glutamic acid and valine.



glutamic acid



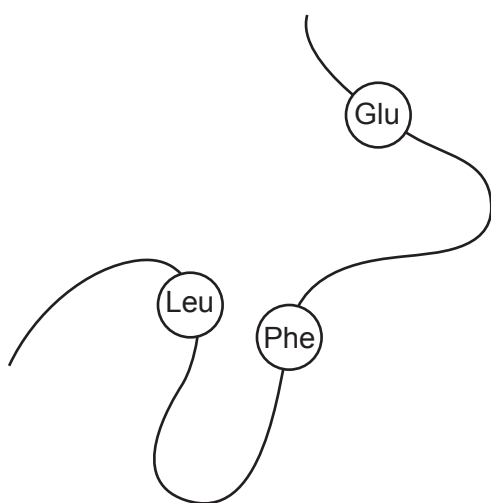
valine

- (i) **Draw a circle** around the R-groups of these **two** amino acids. [1]

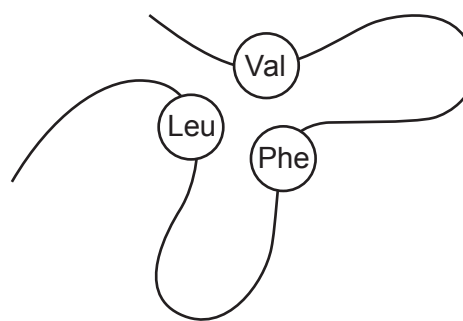
The R-group of glutamic acid is polar whereas the R-group of valine is non-polar. This affects the 3D structure of the resulting beta-globin molecule, as shown in the diagram below.

Key:

Glu – glutamic acid
Val – valine
Leu – leucine
Phe – phenylalanine



Section of beta-globin molecule in a person **without** mutation



Section of beta-globin molecule in a person **with** mutation



- (ii) Valine interacts with phenylalanine and leucine in the beta-globin molecule. Phenylalanine and leucine both have non-polar R-groups. Explain how the properties of these amino acids mean that valine interacts with phenylalanine and leucine but glutamic acid does not. [3]

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